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Dynamics of [Pyr₁₃] [Tf₂N] ionic liquid confined to carbon black

Steffen Merz^a, Peter Jakes^a, Svitlana Taranenko^a, Rüdiger-A. Eichel^{a,b} and Josef Granwehr^{a,c}

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The intrinsic ionic nature of room temperature ionic liquids (RTILs) bears the potential to replace classical aqueous electrolytes in electrochemical applications, for example in metal–air batteries. For a systematic adjustment of RTIL properties in porous cathodes, the ionic arrangement under confinement is of prime importance. Using spectrally resolved pulsed gradient stimulated echo nuclear magnetic resonance (NMR) and spin–lattice NMR relaxation time (T_1) distributions, the dynamics of 1-Methyl-1-propylpyrrolidiniumbis(trifluoromethylsulfonyl)imide ([Pyr₁₃] [Tf₂N]) confined to carbon black were investigated. A considerable dependence of the [Pyr₁₃] mobility on the loading fraction of the carbon black pore space was found. There is evidence for a preferential layering of the RTIL adjacent to the carbon surface and a dependence of the ionic configuration on the local structure of the carbon surface. The inversion efficiency of inversion–recovery T_1 data indicates a quasi-stationary layer at the carbon surface with solid-like properties, where the bulk-like properties of the RTIL are adopted as the distance to the surface increases. From the NMR diffusion data an intermediate layer between the quasi-stationary and the bulk-like RTIL is evident. This layer shows a particularly strong pore space loading dependence. While it has an anisotropic, two-dimensional mobility with reduced diffusion perpendicular to the surface at any loading, when it interfaces a gas phase at low loading its mobility is higher than bulk diffusion by up to an order of magnitude and chemical exchange with other layers is low. This layer appears to be of particular importance for the ion exchange between RTIL environments with different spacing from the carbon surface and hence crucial for the overall dynamics of RTILs in the investigated porous environment.

1. Introduction

In electrochemical systems reactions typically occur at the interface between a solid matrix and a liquid electrolyte or between electron and ion conducting phases. For systems such as metal–air batteries, where a redox reaction between a metal (anode) and the oxygen of ambient air (cathode) drives the electrochemical reaction, a gaseous, insulating phase is further introduced. This leads to an intimate contact between all three phases what is called the three phase zone or triple phase boundary (TPB) [1–3], where the current density for the electrochemical reaction is highest. To increase the TPB, porous gas diffusion electrodes (GDE) are commonly employed and tailored [4, 5]. For electrically rechargeable secondary metal–air systems, GDEs need to possess two major characteristics: i) a high electrocatalytic activity to enable a sufficiently efficient oxygen reduction reaction (ORR) and oxygen evolution reaction (OER), and ii) a sufficiently saturated, but not completely flooded pore space. Therefore, a highly porous structure is crucial for an effective TPB.

To create a large TPB area, the pore space needs to maintain a specific stage of saturation with electrolyte. Understanding the interactions between different phases and quantify the kinetics of the involved processes is crucial for a systematic improvement of materials and electrolytes. In particular for transport-limiting processes it is necessary to address the electrolyte mobility inside a porous electrode.

To offset some of the issues faced with aqueous electrolytes for secondary metal–air systems, such as metal passivation or corrosion, shape changes, dendrite growth, precipitation of insoluble carbonates, water evaporation or ambient moisture uptake [6], non-aqueous alternatives have been introduced [7, 8]. Room temperature ionic liquids (RTIL) represent such an alternative that has been in the focus of research over the last decade [9–12]. RTILs are Coulomb fluids, solely composed of ions, corresponding to salts with melting temperatures T_m below 298K [13]. Their physicochemical properties can be tailored to a particular application by changing the cation/anion configuration and by adding molecular solutes such as water or organic molecules [14]. RTILs can be advantageous over common aqueous alkaline electrolytes since they are liquid over a wide temperature range, of negligible vapor pressure and intrinsically possess a high ionic conductivity.

In RTILs short-range interactions (polarization, π – π , dipole–dipole, solvophobic and van der Waals interactions) act simultaneously with long-range interactions (Coulombic electrostatic forces),

^a Forschungszentrum Juelich, Institute of Energy and Climate Research, Fundamental Electrochemistry (IEK-9), 52425 Juelich, Germany. E-mail: s.merz@fz-juelich.de

^b RWTH Aachen University, Institute of Physical Chemistry, 52074 Aachen, Germany

^c RWTH Aachen University, Institute of Technical and Macromolecular Chemistry, 52074 Aachen, Germany

† Footnotes relating to the title and/or authors should appear here.

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causing complex dynamics in bulk RTILs, at surfaces and under confinement [13, 15-18].

For metal-air systems, RTILs need to possess a sufficiently large electrochemical window, which can be optimized by tailoring the anion/cation composition [19]. The cation size has a significant effect on the viscosity, which in turn influences the TPB as well as the conductivity [19, 20]. Increasing the length of alkyl chains that commonly occur in RTILs can induce ordering of polar groups and changes in ionicity [21-23]. Besides cation/anion composition and the alkyl chain length, the water content of the RTIL or RTIL mixtures is the third major factor determining the properties of RTILs for application in electrochemical systems [24]. Due to their ionic nature, RTILs are subject to strong electrostatic interactions; hence they show a cation/anion dependent dynamic behavior that is distinctly different to molecular liquids. This affects bulk dynamics, where cation-dependent self-aggregation has been found [25], as well as dynamics of IL in a porous host [17, 26, 27], both without [28] and under [29, 30] an applied electric potential. Molecular dynamics (MD) simulations indicated that, *e.g.*, for 1-Ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([emim] [Tf₂N]) confined to a graphitic nanopore slit, ionic mobility is considerably higher parallel than perpendicular to the surface [31]. For RTIL confined to carbon nanotubes, MD simulations suggested an enhanced translational diffusion by up to two orders of magnitude with respect to bulk-phase diffusion due to reduced friction, molecular stacking, and cooperative dynamics effects [32], which has been experimentally confirmed by nuclear magnetic resonance (NMR) [33].

Using quasielastic neutron scattering, Chathoth *et al.* found that for 1-Butyl-3-methylimidazolium-bis(trifluoromethylsulfonyl)imide ([bmim] [Tf₂N]) confined to a nanoporous carbon matrix, cations that are not immobilized near the pore wall show an enhanced mobility compared to the bulk [34]. Later, the same authors found two different mobility regimes, with fast confined and slow bulk RTIL motion for a different RTIL, suggesting that this behavior is somewhat universal and not just valid for a single sample combination. On the other hand, such a trend was not found for 1-Hexyl-3-methylimidazoliumbis(trifluoromethylsulfonyl)imide in silica gel, yet a concentration dependence for the ionic mobility has been reported [35]. There is strong experimental evidence for extended solid-like highly ordered lamellar structures of RTILs under confinement, as shown by atomic force microscopy (AFM) techniques [36-39]. Nevertheless, the origins of the layering are still under debate.

Over the last decade, NMR has proven capable of offering a molecular-based understanding of RTIL dynamics [40-44]. Since rotational, translational and internal motions of molecules are driving relaxation of nuclear spin, studies of longitudinal relaxation time constants T_1 are suitable to characterize the dynamics of molecules in bulk fluids [45], solids [46] and confined fluids [47]. Under confinement, T_1 relaxation and pore dimensions are directly linked [48]. In addition, pulsed field gradient (PFG) NMR methods are often applied to study fluid diffusion [49]. PFG [44] and relaxation [50] NMR studies of bulk RTILs have indicated that

macroscopic dynamics (length scales of micrometers and averaged over timescales of milliseconds to seconds) can be modeled analogous to molecular liquids as unrestricted Gaussian processes, even though MD simulations indicate a different behavior on (sub)-nanosecond timescales [51, 52]. On the other hand, restricted motion of RTILs in porous media showed notable differences such as accelerated motion [33, 53] or diffractive features in disordered porous media [54]. For example, a recent NMR study applying PGSTE NMR on protic ethylammonium nitrate (EAN) confined between polar glass plates found an accelerated diffusion compared to the bulk RTIL [55], consistent with predictions of MD simulations [32].

Accelerated diffusion of RTIL confined to mesoporous systems has also been described in literature, *e.g.* for aprotic [bmim] [Tf₂N] using quasi-elastic neutron scattering [34]. Recently, Galluzzi *et al.* reported on a solid-like insulating lamellar nanostructure for the same RTIL under confinement. These clusters perturb the ionic conductivity, which can influence the performance of the system [56]. Mattea *et al.* investigated [emim] [Tf₂N] and [bmim] [BF₄] in polyvinyl alcohol porous scaffolds [53]. They found an enhanced diffusion for [Tf₂N] under partial saturation compared to the bulk. Nevertheless, all studied solid host materials lack a stochastically disordered morphology intrinsic to carbon materials. Porous carbons are often employed as electrode material in electrochemical devices [57, 58]. Therefore, knowledge about the dynamics of RTIL confined to such basic frameworks containing only carbon is of great importance. To investigate the interactions of porous carbons with electrolytes, NMR is one of the preferred methods [59]. Different environments can be identified in the NMR spectrum from a nucleus-independent chemical shift [60], with ring currents in graphitic carbon as one of the main reasons [61].

In this study we investigated the aprotic RTIL 1-Methyl-1-propylpyrrolidiniumbis(trifluoromethylsulfonyl)imide [Pyr₁₃] [Tf₂N] [62], confined to a porous system consisting solely of carbon black at different loading fractions. Carbon blacks not only possess a complex pore structure with a large inner surface, but in addition a heterogeneous surface [63]. By comparing T_1 relaxation data with PFG diffusion NMR, mobility on different time- and length scales can be investigated. Furthermore, inverting the relaxation and diffusion data allows us to further employ the spectral dimension, which is intrinsic in the measured free induction decay or echo, to identify different environments in the porous host and to obtain a fairly detailed model of the loading and dynamics of the RTIL in the carbon black matrix.

2. Methods

Spectrally resolved T_1 relaxation

Longitudinal relaxation, also called spin-lattice or T_1 relaxation, describes the exchange of energy of a spin system with its environment that eventually leads to a thermal equilibration. It is caused by fluctuating interactions with a non-vanishing spectral density at the Larmor frequency of the nuclei under investigation. Thus, T_1 is a measure of the average lifetime of nuclei in a non-equilibrium state.

For liquid-state samples, T_1 relaxation times are commonly measured using the inversion–recovery (IR) method [64]. First, a 180° (π) pulse inverts the spin magnetization from equilibrium. After a recovery time τ , a 90° ($\pi/2$)_x detection pulse is applied in order to transfer longitudinal magnetization into detectable $-y$ magnetization that is measured in the form of a free induction decay (FID). The FID is proportional to the longitudinal magnetization at the time of the ($\pi/2$)_x pulse. Fourier transform of this FID provides the instantaneous NMR spectrum of the excited nuclei. By varying τ , the return of the longitudinal magnetization to equilibrium is sampled point-by-point, and T_1 may be obtained by fitting the time constant of an exponential function. In the presence of chemical exchange between different environments (magnetic site exchange), T_1 will be shortened and an unambiguous assignment of relaxation and exchange time constants is generally not straightforward [65]. For a simple two-site exchange model system, this is demonstrated in the Supplementary Information (SI–4). On the other hand, exchange rates larger than T_1 have generally a minor effect on the NMR signal.

Pulsed field gradient diffusion NMR

One fundamental element in the dynamics of liquids in bulk or confined to porous matrices is their self-diffusion. PFG NMR diffusion measurements label the spatial position of nuclei by assigning their transverse spin magnetization a location-dependent phase, which is refocused to an echo after a mixing time Δ . The signal attenuation from incomplete echo formation due to incoherent translational displacements over a time period Δ is a measure of the mean displacement of the observed nuclei, from which the molecular diffusion coefficients can be deduced. The pulsed gradient stimulated echo (PGSTE) pulse sequence, $\pi/2 - \delta - \pi/2 - \Delta - \pi/2 - \delta - \text{echo}$, where δ is the time during which a constant magnetic field gradient with amplitude g is applied across the sample, is a common method to measure diffusion processes. For molecular liquids, PFG NMR diffusion experiments are commonly interpreted by assuming Brownian random motion of the molecules in bulk or under confinement [66].

Laplace inversion without non-negativity constraint

IR and PGSTE NMR data can be converted into a continuous distribution of relaxation or diffusion components, respectively, using a Laplace inversion with a suitable kernel [67]. The inversion is numerically ill-conditioned for the exponential kernel that is usually used to analyze relaxation and diffusion data. The result can be stabilized by regularization. For NMR application, Tikhonov regularization in combination with a non-negativity (NN) constraint is a popular method to prevent oscillating solutions. NN assumes that all relaxation components have an identical sign. This approach does not hold for exchanging systems, since components with negative signs, which are inherently possible in systems with chemical exchange, will be canceled, leading to artificial features in the obtained density function [68] (see SI–4).

3. Experimental

Sample preparation

Vulcan carbon black XC 72 powder (Cabot Corporation, U.S) with a BET surface area $254 \text{ m}^2 \text{ g}^{-1}$ and average primary particle size of 30 nm was used as pellet material. 1-Methyl-1-propylpyrrolidiniumbis(trifluoromethylsulfonyl)imide ([Pyr₁₃] [Tf₂N]) was purchased from Iolitec (Germany) and used as received (water content < 500ppm, density = 1.35 g cm^{-3} at 293,15K). Pellets of 2 mm thickness were prepared from 60 mg Vulcan XC 72 carbon black powder using an 11 mm inner diameter circular shaping die with a uniaxial press (P/O/Weber, Germany) and a forming pressure of 85 kN for 5 minutes. The pellet was placed into a 25 mm outer diameter glass test tube for NMR imaging (Hilgenberg, Germany). [Pyr₁₃] [Tf₂N] was added using a micro syringe. Experiments were started after equilibrium was achieved (all ionic liquid was adsorbed by the pellet). An amount of 133 mm^3 [Pyr₁₃] [Tf₂N] was applied to fully saturate the pellet ($c=1$), and 53 mm^3 and 17 mm^3 for a pellet saturation of $c \sim 1/3$ and $c \sim 1/10$, respectively. The mass added was determined by weighing the system before and after RTIL application.

NMR experiments

All ^1H -NMR experiments were performed using a 14.1T Bruker Avance III spectrometer equipped with a 1.5 T m^{-1} gradient system (Micro 2.5) and an imaging probe (MICWB40) with a 25mm inner diameter ^1H quadrature coil. A probe temperature of 291,15K was maintained during the experiments. A background spectrum was recorded and subtracted for each measurement. The stimulated echo version of a PGSTE experiment, following the approach of Stejskal and Tanner [69], was applied using a gradient pulse duration $\delta = 6 \text{ ms}$, time between two gradient pulses $\Delta = 150 \text{ ms}$, 256 gradient steps, recycle delay $\text{RD} = 3 \text{ s}$, a radio frequency pulse duration of $\tau_{90} = 0.022 \text{ ms}$ for 90° pulses and a pre-scan delay = 0.0065 ms . Longitudinal relaxation time constant (T_1) measurements were performed using an inversion recovery (IR) sequence with $\text{RD} = 11 \text{ s}$, spectral width $\text{SW} = 125 \text{ kHz}$ and a variable delay list with recovery delays between $10 \mu\text{s}$ and 10 s .

Data analysis

For fluids undergoing Brownian motion, the attenuation of the echo amplitude is given by the Stejskal–Tanner relation [69],

$$\frac{S(g)}{S(0)} = \exp \left(-\gamma^2 g^2 \delta^2 D \left(\Delta - \frac{\delta}{3} \right) \right) \quad (1)$$

where S is the signal intensity, γ is the gyromagnetic ratio ($2.6751 \times 10^8 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}$ for ^1H), g and δ are the amplitude and duration of the gradient pulse, Δ is the interval between the two gradient pulses and D the self-diffusion coefficient [70–72]. Both the diffusion and the T_1 data were spectrally resolved by Fourier transforming the NMR raw data along the transient dimension. The spectral dimension was referenced vs. TMS using published chemical shifts of the neat RTIL [62]. The distributions of T_1 and of the diffusion coefficients were obtained by performing a regularized inversion with an exponential kernel without non-negativity constraint [73, 74]. Parameterization was done as described in [73]

without manual fine-tuning the parameters (see SI–1 for a complete parameter list). In the figures, the T_1 and the diffusion coefficient distributions are represented as square root scaled color maps with conservation of the sign to enhance weaker spectral features.

4. Results

RTIL loading dependent NMR spectra

First, only the NMR spectral dimension of the 2D frequency resolved relaxation and diffusion experiments is discussed to highlight relaxation- or diffusion-independent features. IR and PGSTE detected spectra show different T_2 contrast. A comparison of the two detection modalities facilitates a distinction between spectral features with i) very short T_2 on the order of the duration of the IR inversion pulse, which cannot be fully inverted in the IR experiment, ii) features with short T_2 for which $T_2 \ll \delta$, which can be inverted in the IR experiment but do not show a signal in the PGSTE experiment, and iii) features with long T_2 that show a signal in the IR as well as in the PGSTE experiment.

The ^1H NMR spectra of $[\text{Pyr}_{13}]$ for different RTIL loading of the pore space, obtained from the IR T_1 experiments with the longest recovery delay and from PGSTE diffusion experiments with the weakest gradient, are shown in Figure 1a. Confining the RTIL to the carbon black matrix caused a widening of the spectral features. Individual resonances broadened and often became indistinguishable since the ions experience a distribution of magnetic fields caused either by susceptibility differences between the carbon matrix and the RTIL, by local ring currents in graphitic parts of the carbon surface, or by local magnetic fields induced by currents in the electrically conductive matrix [61]. This is consistent with observations of molecular liquids in porous media, whose NMR spectra are broadened due to the presence of susceptibility gradients between a porous host and fluid in its void space [75].

At a loading of 10% of the pore space no PGSTE detected ^1H NMR spectrum could be recorded, hence only species with short or very short T_2 relaxation time constant, located most probably in the immediate vicinity of the carbonaceous surface, are contributing. In other words, surface or near-surface environments with short T_2 get filled first upon loading of the porous carbonaceous host with RTIL. In the IR detected spectrum of the sample with 10% pore space loading, two modes separated by approximately 13 ppm can be distinguished. This observation at first appears to be consistent with reports in [28] where the downfield resonance has been interpreted as originating from more bulk-like cations in intergranular pores ("ex-pore"), and the upfield resonance as cations in micropores ("in-pore"), experiencing a ring current effect from aromatic parts of the carbon matrix [46]. However, care has to be taken when adopting this interpretation, which was based on data acquired under different experimental conditions, namely by magic angle spinning (MAS) of the sample. In addition, VXC-72 shows a wide range of pore sizes [76] with diameters that preclude an in-pore interpretation for a significant fraction of the sample. Furthermore, the chemical shift (CS) difference of about 13 ppm between the two modes is almost twice the value reported in [28], and the downfield-shifted peak is deshielded compared to the bulk

resonances of the RTIL. Such a CS difference would be consistent with CS differences expected between downfield-shifted nuclei located in the plane of an aromatic ring and upfield-shifted nuclei above an aromatic ring system [77]. Schröder *et al.* [63] distinguished four different surface types for carbon black: (i) graphitic planes, (ii) amorphous carbon, (iii) edges/surfaces of graphitic crystallites, and (iv) slit-shaped cavities formed by adjacent graphitic crystallites. RTIL located on surfaces of type (iii) or type (iv) would directly experience a downfield shift consistent with the observed value. Alternatively, regions on the carbon surface with different conductivity and therefore different ionophilicity, such as type (ii) amorphous carbon, may not directly induce a downfield shift, but instead could cause a different structure for the surface RTIL [56]. For example, a downfield shift for $[\text{Pyr}_{13}]$ could be induced by an anisotropic magnetic shielding effect caused by a partial ordering of the $[\text{Tf}_2\text{N}]$ anions [78].

Notice that the downfield mode overlaps with the mode expected for bulk RTIL. Hence an unambiguous assignment of resonances is not possible from spectral data alone. Nonetheless, some further distinction between different environments is possible by addition of the IR spectra with the shortest and the longest recovery delay, shown in Figure 1b for the three samples. Fully inverted components, which show equal amplitudes with opposite sign, will subtract, while components with very short T_2 or which experience exchange at an intermediate rate show up in such a sum spectrum. This also helps identifying spectral components that are not quantitative in the IR experiment, since an incomplete inversion with recovery time $\tau = 10 \mu\text{s}$ also implies that a $\pi/2$ pulse does not quantitatively flip longitudinal into transverse magnetization. For the sample with 10% pore space loading, the upfield component can barely be inverted, while the downfield peak gets almost fully inverted. This implies faster T_2 for the upfield component, which is probably caused by a lower mobility of the responsible nuclei. It also implies that the upfield component is more intensive than suggested by its amplitude.

At a loading of $1/3^{\text{rd}}$ of the pore space primarily the downfield mode of the IR detected ^1H NMR spectrum intensifies, indicating that the environment responsible for the upfield mode is preferentially loaded and already saturated in the $1/10^{\text{th}}$ sample. Once it is full, the environment responsible for the downfield mode gets filled. The sum spectrum of IR spectra with shortest and longest recovery delay also show little change for the upfield peak, while the downfield peak gets considerably more intense. In-between the two boundary regions, where the resonances of bulk-like RTIL would be expected, a negative signal is obtained. This may be driven by chemical exchange taking place between the different environments.

In addition, the bimodal character of the spectrum is less pronounced compared to the sample with lower pore space loading. Only the upfield part of the shielded surface mode remains separated, whereas the downfield part starts to merge with the deshielded mode from bulk-like RTIL. Such an effect could be caused by a loading dependent exchange between the two environments, with an increased rate at higher pore space loading.

The 1/3rd loaded sample also shows a PGSTE detected spectrum, yet with an amplitude that amounts to only a small fraction of the total number of ¹H nuclei in the sample. Its position is not located at the position of the bulk-like RTIL but in-between the two modes with a slight tendency towards the upfield peak. This implies that the nuclei responsible for the signal are still influenced by the graphitic surface, as this is the only CS contribution causing such a strong shielding in this sample, but that they are some distance away from the surface [61]. The formation of a more mobile layer, causing a longer T_2 and hence PGSTE detectability, on top of the almost stationary layer on the graphitic part of the carbon surface, would explain such a signal. More unusually, the PGSTE detected spectrum shows individually resolved resonances for the different ¹H nuclei. This is only possible if the nuclei remain at the same distance from the surface for a prolonged amount of time, while at the same time anisotropic broadening caused by the statistical orientation of the carbon surface with respect to the external magnetic field B_0 is averaged out.

At full loading of the pore space the NMR spectrum from the IR data shows a single mode without considerable fine structure. This mode is located in-between the upfield and the downfield modes in the spectra from the partially loaded samples, indicating a further increase of the exchange rate between the different environments. Such an increased exchange rate is supported by the observation that at the position of the upfield mode at reduced loading, the signal amplitude is reduced in absolute terms, not just relative to the overall signal, which points towards coalescence of the NMR signals from different environments [79].

The PGSTE spectrum of the fully loaded sample shows qualitatively different features compared to the spectrum of the 1/3rd loaded sample. The position of the signal is shifted downfield to the position expected for bulk-like RTIL, close to the center of the IR detected spectrum. In addition, the resolution of the PGSTE detected spectrum is reduced. The individual resonances are not resolved any more, yet the overall width is still similar to the partially loaded spectrum and therefore narrower than the IR detected spectrum.

The PGSTE-detected signal increased by considerably more than the factor three that would be expected from the additional amount of RTIL in the sample. This is consistent with the assumption that larger, more bulk-like pores with reduced impact on T_2 get filled at the end. According to the signal amplitude, still not all of the cations, yet a larger fraction than for the 1/3rd loaded sample, are contributing to the PGSTE spectrum. Hence even for the fully loaded sample a considerable fraction of the signal in the IR spectrum is caused by cations with $T_2 \ll \delta$ that may be located near the carbon surface for a prolonged amount of time that prevents an increase of T_2 [80].

Diffusion of confined [Pyr₁₃][Tf₂N]

Figure 2 shows the spectrally resolved distribution of the apparent diffusion coefficients of the [Pyr₁₃] cations for the saturated carbon matrix and a loading factor of 1/3rd. The self-diffusion coefficient, D_0 , of [Pyr₁₃] in the neat RTIL ($D_0 = 7.5 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$) is indicated by a white

line and corresponds to the diffusion coefficient reported in [81]. As can be seen in the projection of the diffusion coefficient (Fig. 2a, side panel), which is linearly scaled and therefore the signal is proportional to the number of contributing nuclei, the main mode is only weakly, but noticeably affected by the porous host. In particular, the maximum of the distribution is located at $D \approx 6.9 (2) \times 10^{-12} \text{ m}^2/\text{s}$ for the fully loaded sample and at $D = 8.9(2) \times 10^{-12} \text{ m}^2/\text{s}$ for the 1/3rd loaded sample.

For the sample with 1/3rd of the pore space loaded with RTIL (Fig. 2a) the resolution of the different ¹H resonances discussed above is clearly visible also in the 2D map. PFG-based diffusion experiments measure long-range mobility of the detected nuclei over a duration set by the diffusional mixing time Δ . Since only the highly mobile layer on top of the low mobility surface layer, which was identified from the IR detected spectra, shows T_2 values long enough for PFG detection, the corresponding nuclei experience the same environment for a prolonged time on the order of Δ . Therefore exchange between this layer and the low mobility surface layer must be slow.

In addition, a correlation between the effective diffusion coefficient distribution and the frequency is observed. The upfield part of each resonance, which is shifted about -5 ppm from the bulk-like position, shows an effective diffusion coefficient D at about the D_0 value of [Pyr₁₃] or slightly below. Farther downfield, each of the resonances shows acceleration, with D larger than D_0 , implying that displacement due to random motion increases above the value of the bulk RTIL. At the downfield end of each resonance the diffusion accelerates considerably, with D showing a broad distribution with its center about an order of magnitude higher than D_0 . At the same time, the upfield part of each resonance shows a negative contribution at the same high effective D values, which indicates at least some exchange between the different environments.

The resonance frequency, which is shifted upfield compared to bulk-like RTIL, indicates that this highly mobile layer is located exclusively on top of the low mobility RTIL layer above graphitic regions. If it would also move over the RTIL on amorphous regions or crystallite edges of the carbon black, which show a downfield shift, it is unlikely that the resolution of individual resonances could be maintained.

To obtain an idea of the distances sampled by [Pyr₁₃] in a particular layer without mixing, the relationship $\sqrt{\langle r^2 \rangle} = \sqrt{2D_0\Delta}$ can be used, where r is the length scale sampled by an investigated species. When using the bulk self-diffusion coefficient of [Pyr₁₃] and a mixing time $\Delta = 150$ ms, then a value of $r \approx 1.5 \text{ }\mu\text{m}$ is obtained. This large lengthscale in combination with the above-mentioned confinement of the mobile RTIL to graphitic regions of the carbon black implies that the graphitic surface regions are, overall, well connected and that the amorphous carbonaceous regions or graphitic crystallite edges form islands within the graphitic surface. Nonetheless, according to the amplitude of the downfield shifted resonance, these islands cover a fairly substantial fraction of the overall surface. The chemical shift variations within the mobile fraction may have by a number of different reasons [82]. It could be a pore size effect, a dependence on the domain size and curvature of the aromatic

carbons, or a function of the distance to the boundary of graphitic domains. Furthermore, it may be reflecting the mean distance of the nuclei from the surface of the carbon matrix [61]. It is fairly unlikely that the domain size is at the origin of this effect, since it would imply domain size selective motion over micrometer lengthscales. In addition, a constant distance from the boundary of the graphitic domains would contradict the formation of islands by the amorphous regions, in which case no long-range motion with constant distance would be expected. The other explanations are therefore more likely and may in fact be linked. The magnitude of the chemical shift variation of each resonance by about 1 ppm is consistent with variations predicted due to curved graphitic domains, with lower curvature showing a smaller upfield shift [28]. The curvature of such aromatic domains may be related to the pore size and, besides, the distance of the [Pyr₁₃] cations may also be related to the curvature. It is thus likely that the RTIL moves somewhat pore lengthscale selective, which implies a certain hierarchical porosity in the solid host. The strong exchange features for the very fast moving contributions further imply that [Pyr₁₃] located near the surface of large pores exchanges most readily with IL in smaller pores. This layer is therefore crucial for the global mobility of RTIL within the investigated carbon host.

In the spectrally resolved distribution of the apparent diffusion coefficients of the RTIL saturated sample, one broad mode was observable with a main intensity slightly below the diffusion coefficient of the neat RTIL (Fig. 2b). The spectral resolution was lost, yet a weak feature with accelerated effective diffusion remained. This feature was bipolar, with positive sign on the downfield wing and negative sign on the upfield wing of the spectrum, which indicates exchange with a time constant on the order of Δ . At the same time, the features at the position of the 1/3rd loaded sample had disappeared, which implies that the very low longitudinal mixing of the highly mobile layer on top of the immobile surface RTIL only exists when facing a gas phase, or maybe an immiscible phase, on the opposite surface. If this volume, wherein RTILs behave more bulk-like, is filled then the high mobility still persists, but exchange with the bulk-like phase prevents the same long-time stability of the mobile layer that was observed with partial loading of the pore space. On the time scale of Δ , therefore, a gradual decrease of the RTIL mobility as a function of distance from the surface is observed, eventually reaching the D_0 value of the neat RTIL for cations far away from the nearest surface. This is consistent with results in [83], where it was suggested that the increased diffusion at low loading fractions can be understood as transition from a 3D-like (saturated pores) to a 2D-like structure (only a film or layer coating the porous surface), which affects the ion dynamics dictated by Coulombic interactions.

T_1 relaxation time distributions

Figure 3 shows the spectrally resolved ^1H T_1 relaxation time distribution of [Pyr₁₃] at different loadings of the carbon pellet. The T_1 distribution is dominated by a single relaxation mode at about $T_1 \approx 500\text{ms}$ for all pore saturation stages of the pellet, which is, within the width of the relaxation mode, equal to the proton T_1 of

the neat RTIL. Only the upfield-shifted resonance shows a slight T_1 increase of this mode to about 1 s, indicating a somewhat altered local mobility of the respective spins in combination with incomplete exchange mixing on the timescale of T_1 .

In addition, substantially weaker saturation-level dependent relaxation features with considerably shorter time constants are observed (see SI-2 for an analysis of the residuals). These features are characteristic for exchange processes with a slow-to-intermediate rate, *i.e.* with an exchange rate constant k_{ex} on the same order of magnitude as the frequency difference $\Delta\nu$ between the exchanging species or slightly below. Exchange is further supported by the presence of dispersive features at short T_1 values along the frequency dimension [84], leading to components with negative sign in the relaxation distribution of all three investigated samples (see SI-2 for the significance of the negative components and SI-4 for a visualization of the described effect). In this intermediate regime, a change of the exchange rate also leads to a shift of the resonance frequencies of the exchanging components. The faster this exchange, the closer the resonances of the involved species move together [79]. The investigated samples show a trend towards smaller frequency differences between exchange features with increasing pore space filling, which shows that mixing of RTIL in the different environments becomes more efficient at higher pore filling levels.

The absence of multiple modes with substantial amplitude in the relaxation dimension of the 2D data implies that the different ^1H nuclei in the [Pyr₁₃] cation all show very similar or even identical T_1 values. Given the differences reported in literature from diffusion ordered spectroscopy (DOSY) experiments of neat RTILs, the good signal-to-noise ratio for the fully loaded sample would suggest that such differences should be resolvable in the present samples as well. Furthermore, the strong variation of local RTIL environments in the porous carbon black (*e.g.* surface vs. bulk-like) implies that substantially differing T_1 should be measured. In solid-state samples, it is often observed that spin diffusion leads to an averaging of T_1 for overlapping resonances. The same mechanism has been suggested for imidazolium-based RTILs at temperatures close to their melting point [42]. Such an explanation would also satisfy the current observations. With a melting point of 12 °C for [Pyr₁₃] [Tf₂N], it is plausible that the sample temperature of 18 °C is still within the "low-temperature region" specified in [42], in particular since for surface species, ion dynamics is slowed down with an effect similar to a temperature reduction. In addition, Han *et al.* suggested for the N-butyl-N-methylpiperidinium cation that the rotational correlation time of the methyl group connected to the nitrogen shows a hindered rotation with a correlation time similar to the overall molecular reorientation time of the cation in bulk RTIL. For the corresponding pyrrolidinium cation the results were similar [50]. If it is assumed that the N-CH₃ rotation is largely independent of the local [Pyr₁₃] environment, such a model could explain the low pore loading dependence of T_1 .

While exchange processes, possibly in combination with spin diffusion, may average out the mean T_1 value measured experimentally, the changing probability for the different

environments in the samples with varying pore space loading suggests that the mean T_1 should change between the samples. Therefore, the observed similarity of this time constant for neat RTIL and RTIL in the porous medium indicates that, despite a potentially reduced translational mobility, the local mobility process primarily responsible for T_1 relaxation is affected only weakly for most of the RTIL by the porous host.

Since the samples are in a dynamic equilibrium, no net redistribution of cations during the experiment occurs, therefore exchange only leads to observable features in the 2D T_1 vs. frequency data if nuclei in different environments experience different T_1 relaxation time constants or if inversion is incomplete for some of the features. Assuming an averaging of T_1 by spin diffusion, the observed features at short T_1 would have to be linked to exchange processes with nuclei experiencing incompletely inverted magnetization by the preparation pulse of the IR sequence. As can be seen most clearly in the projections onto the relaxation axis in Figure 3, the signal contributions at short T_1 are small enough to be consistent with this explanation. While it is generally challenging to distinguish between relaxation and exchange in particular for ordered or partially ordered samples [85] this suggests that the observed short time constants could be directly interpreted as exchange time constants.

5. Discussion

Figure 4 schematically sketches the various RTIL environments for different loading fractions of the carbon pellet. At low loading of the pore space, a low mobility layer forms directly on the surface (Fig. 4d green). This layer preferably forms on surface regions of the carbon black consisting of graphitic planes that show the highest ionophilicity, which can be recognized from an upfield shift of the ^1H resonances of [Pyr₁₃]. With a reduced affinity, RTIL is deposited on less ionophilic surface regions such as amorphous carbon regions or on graphite crystallite edges and slit-shaped cavities of the carbon black, which show a resonance that is downfield shifted compared to the bulk spectrum (Fig. 4d blue and teal).

Such a model would not be consistent with an "in-pore" interpretation of the upfield mode as suggested for this RTIL based on different NMR experiments using a different type of carbon black [61]. Instead, due to the ionophilic surface of the carbon and the high hydrophobicity of both the RTIL and the carbon surface, it is expected that the surface gets fully covered with a thin film, including the filling of small yet accessible pores, before larger, more bulk-like voids start to be filled.

While the reservoir for RTIL on graphitic planes saturates quickly, on surface regions with a different morphology the saturation appears to be more gradual and concurrent with the formation of additional layers on the graphitic surfaces. At a pore space loading of $1/3^{\text{rd}}$, also a PFG-detected signal with a high effective diffusion constant appears. Its resonance frequency is located between the signal from RTIL on graphitic planes and bulk-like RTIL. This shift indicates the formation of a double layer on the graphitic planes, with the

high mobility cations selectively located above the low mobility layer on graphitic planes (Fig. 4d red). Notice that a long-range mobility of such a layer, as probed in a PFG experiment, only works once the ionophilic graphitic planes have been fully covered with RTIL to form an interconnected layer, since disruptions of the layer would also interrupt the long-range motion of ions.

The selectivity of the highly mobile layer to layers located on graphitic planes suggests that the ion configurations in the quasi-immobile layers above the different carbon surface structures must be substantially different; otherwise such a carbon surface discrimination of the second layer would not be expected. It is known that besides the polar head group, the apolar alkyl chains in RTIL's can affect the nanostructural organization driven by van der Waals interactions [17]. There is evidence for a temperature influence on the van der Waals interactions of the alkyl side chains in pyrrolidinium-based RTIL, with increased interactions and thus additional ordering effects driven by the RTIL side chains towards lower temperatures [22]. The decrease in the dimensionality of the ion mobility from 3D-like to 2D-like may have a similar effect as a temperature decrease, thus the van der Waals driven ordering of the alkyl chains could lead to an additional immobilization of the RTIL layer at the carbon surface, which may be strongly dependent on the properties of the carbon surface. Although there is a significant dependence on the alkyl side chains length in pyrrolidinium based RTIL [86] it may be fundamentally different under confinement [17]. There is evidence that the system structure changes from a dispersed microphase to a continuous microphase towards prolonged alkyl chains $\text{C}_n\text{H}_{2n-1}$ for $n \geq 3$ [17]. Although this has only been reported for neat RTIL's it may also contribute to ion orientation and dynamics under confinement.

The ^1H resonances of [Pyr₁₃] in the PFG-detected data show a spectral resolution that is unusually high for a fluid in a porous medium. Such an averaging out of inhomogeneous interactions is only possible if all the detected nuclei statistically sample the full space occupied by the fluid, which is not generally observed for molecular liquids. Since RTILs show a fairly high viscosity and a lower D_0 than most molecular solvents, such an averaging is only realistic if the distance of the cations to the carbon surface remains essentially constant during the diffusional mixing time Δ . Furthermore, the mobility must be high enough to uniformly sample the different orientations of the surface with respect to B_0 in a time that is short compared to the inverse of the inhomogeneous line broadening in a static configuration. This implies the formation of a layer with a very high 2D mobility parallel to the carbon surface, with cations experiencing a very slow exchange with the other RTIL reservoirs at $1/3^{\text{rd}}$ pore space loading (Fig. 4d red) [87].

At low pore space loading, a slow exchange is observed between different environments of the RTIL. When increasing the pore size loading, the mixing between the different environments increases, eventually causing coalescence of the resonance modes. A better contact between RTIL in different adjacent environments as each of them gets filled may be an explanation. While some difference of the spectral signature and the amplitude between IR-detected and

PFG-detected NMR spectra persists, exchange narrowing at full pore space loading indicates that the permeability between layers increases once a bulk-like phase exists instead of an RTIL-to-air interface after a few layers.

Considering the inversion efficiency of the preparation pulse in the IR experiment for different pore space loadings (Fig. 1b), the upfield mode assigned to RTIL on graphitic planes is less affected by very fast T_2 at higher pore space loading, indicating that the respective cations exchange faster with the emerging bulk-like phase than the cations responsible for the downfield surface mode. This could indicate that the structure the RTIL assumes on the graphitic planes is more similar to the bulk structure; hence less activation energy is required to exchange [Pyr₁₃] cations between RTIL on graphitic plane surface with bulk than between RTIL on amorphous carbon or crystallite edges and slit-shaped cavities with bulk.

Based on the frequency difference $\Delta\nu$ (with units of Hz) between exchanging signal components in the slow exchange limit, the exchange rate

$$(2) \quad k_{\text{coal}} = \pi \frac{|\Delta\nu|}{\sqrt{2}}$$

at which coalescence would occur represents an upper limit for the actual exchange rate k_{ex} , as long as coalescence does not actually occur. Using the frequency difference between the modes in the samples with 1/10th pore space loading, $\Delta\nu = 7.5$ kHz, a minimum mean exchange time of $\tau_{\text{crit}} = 1/k_{\text{coal}} \approx 0.06$ ms was calculated for [Pyr₁₃]. This is consistent with rates obtained from the T_1 distribution, where the fastest contributions appear at around $k_{\text{ex}} \approx 10^4 \text{ s}^{-1}$ (Fig. 3a, inset side panel), if fast relaxation components are interpreted as exchange rates. At 1/3rd pore space loading, coalescence commences but is not yet complete. A prominent mode in the T_1 distribution is centered at $k_{\text{ex}} \approx 10^{4.6} \text{ s}^{-1}$, which is beyond k_{coal} . However, this mode extends below $k_{\text{ex}} \approx 10^4 \text{ s}^{-1}$, where coalescence does not set in (Fig. 3a, inset side panel). Finally, at full pore space loading, the spectral features are dominated by coalescence of the lines, and in the T_1 distribution a relatively narrow mode is visible at $k_{\text{ex}} \approx 10^{4.9} \text{ s}^{-1}$.

6. Conclusions

We reported on the dynamics of different pore loading fractions of [Pyr₁₃] [Tf₂N] confined to a carbon black matrix using ¹H PGSTE diffusion NMR and T_1 relaxation time spectra. There is evidence for a quasi-stationary RTIL layer or film that coats the carbon surface. This immobile layer has a lubricating effect on the layer directly above that shows an increased long-range mobility where the anisotropy of ring current effects, line broadening due to susceptibility gradients and dipole–dipole interactions are averaged out. This behavior causes a faster overall motion along the carbon surface in the porous medium than in the bulk RTIL. On the other hand, higher RTIL loading causes faster local exchange between pores or RTIL phases, but the global motion slows down since the residence time in RTIL layers with very high mobility is reduced.

In summary, it was demonstrated that pore space loading has the potential to qualitatively change the RTIL dynamics in environments relevant for electrochemical applications. For an integral description of confined RTIL mobility, long-range diffusion must be considered separately. PFG NMR experiments are in principle suitable for such a characterization, but one must be aware that signal components with short T_2 are not visible in the respective data; hence a quantitative description requires additional experiments that are also able to address very fast relaxing species.

Conflicts of interest

There are no conflicts to declare.

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Figures

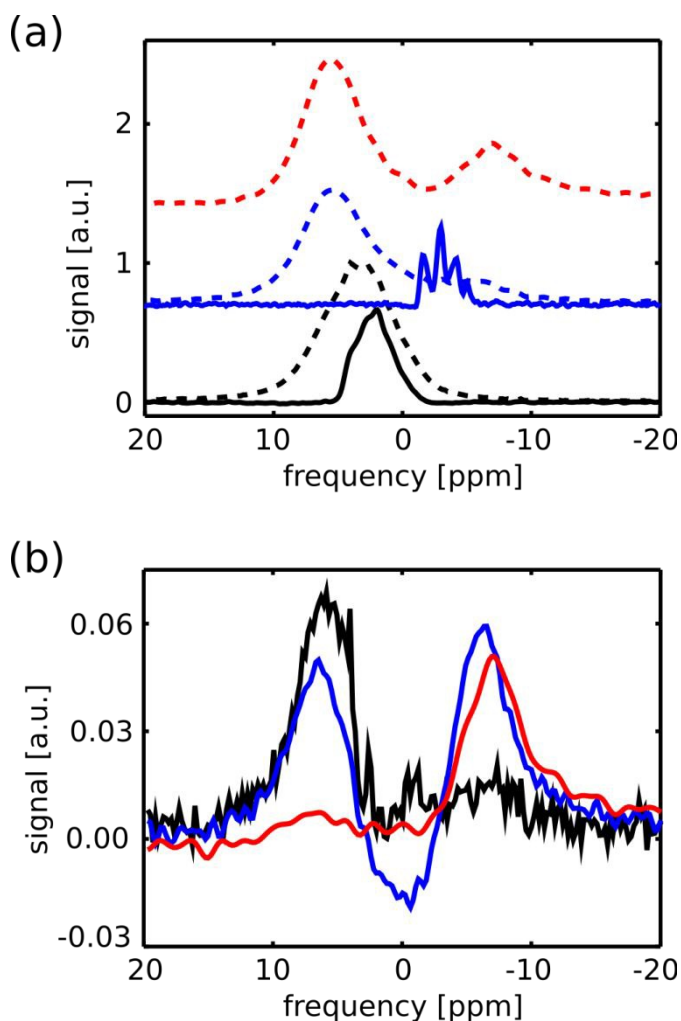
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Figure 1: ^1H NMR spectra of $[\text{Pyr}_{13}][\text{Tf}_2\text{N}]$ in porous carbon black with $1/10^{\text{th}}$ (red), $1/3^{\text{rd}}$ (blue) and all (black) of the pore space loaded with the RTIL. (a) Spectra obtained from the IR experiment with the longest recovery delay (dashed lines) and from the PGSTE experiment with the lowest PFG amplitude (solid lines). The amplitudes of the spectra are divided by the pore space loading fraction. Furthermore, the amplitude of the PGSTE detected spectrum of the $1/3^{\text{rd}}$ loaded sample (solid blue) is scaled by an additional factor 25 and the spectrum of the fully loaded sample (solid black) by an additional factor 5. For the $1/10^{\text{th}}$ loaded sample the PGSTE experiment yielded no signal. (b) Sum spectra of the IR detected spectra with longest and shortest recovery delay, showing signal contributions that could not be fully inverted by the inversion pulse. The signal amplitude is unscaled. The noise levels vary since a different number of data points was used for processing, motivated by the different signal-to-noise ratios of the data sets. All data sets were then zero-filled to the same size before Fourier transform.

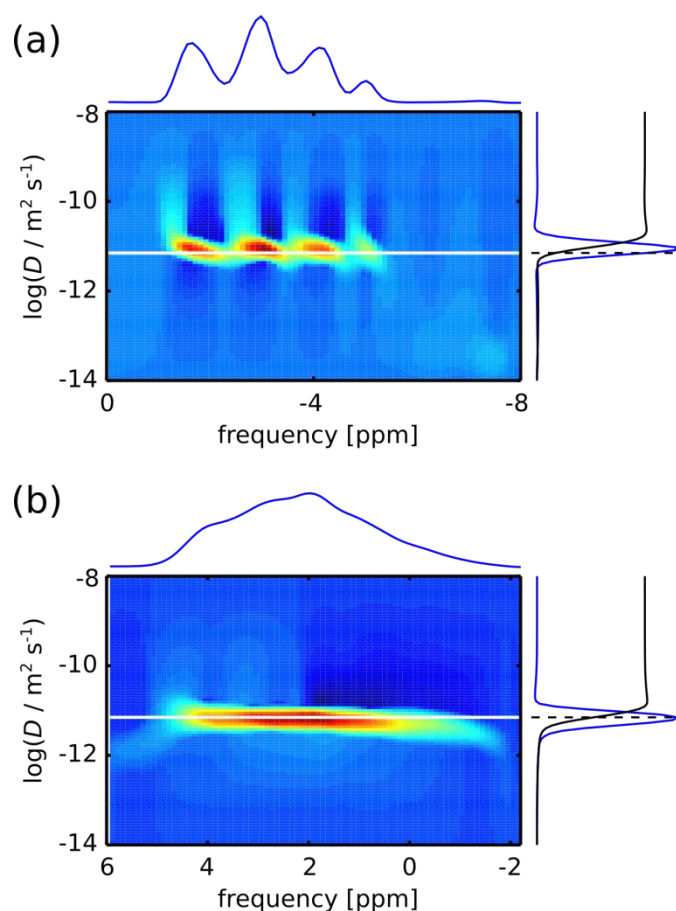


Figure 2: Pseudocolor plot of $[\text{Pyr}_{13}]$ diffusion coefficient distribution vs. ^1H NMR frequency for (a) the sample with $1/3^{\text{rd}}$ of the carbon black pore space loaded with $[\text{Pyr}_{13}] [\text{Tf}_2\text{N}]$, and (b) the sample with fully loaded pore space. The colormap is scaled with the square root of the amplitude to amplify features with low intensity. The sum projection onto the frequency axis is shown on top of each panel and the sum projection onto the diffusion axis (blue) and the cumulative sum (black) to its right. The line (white in the main panel and dashed black on the side) represents the self-diffusion coefficient of the neat RTIL. Notice that the two panels show a different spectral region.

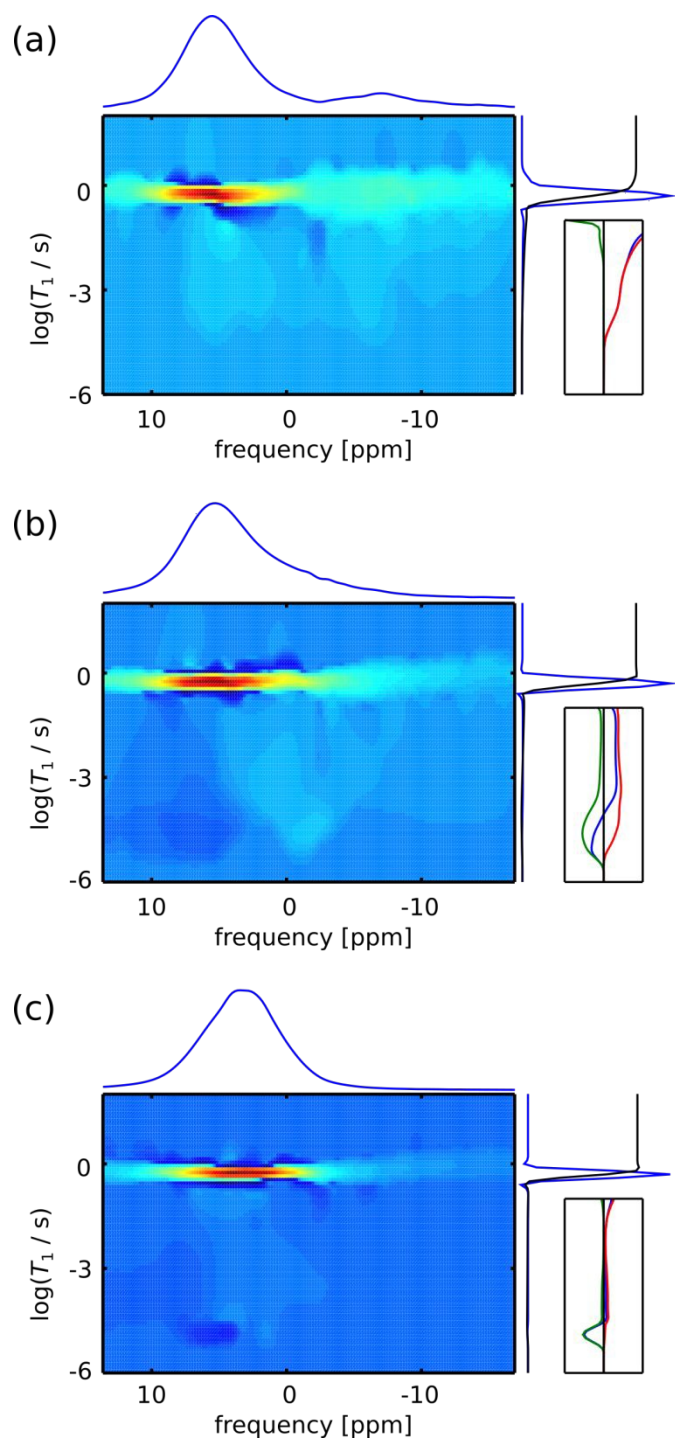


Figure 3: Pseudocolor plot of $[\text{Pyr}_{13}]^+$ T_1 relaxation time constant distribution vs. ^1H NMR frequency for (a) the sample with $1/10^{\text{th}}$ of the carbon black pore space loaded with $[\text{Pyr}_{13}]^+ [\text{Tf}_2\text{N}]^-$, (b) the sample with $1/3^{\text{rd}}$ of pore space loaded, and (c) the sample with fully loaded pore space. The colormap is scaled with the square root of the amplitude. The sum projection onto the frequency axis is shown on top of each panel and the sum projection onto the relaxation axis (blue) and the cumulative sum (black) to its right. The inset shows the region with short T_1 scaled by a factor 35 (blue) along with the sum of all the components in the distribution with positive (red) and with negative sign (green).

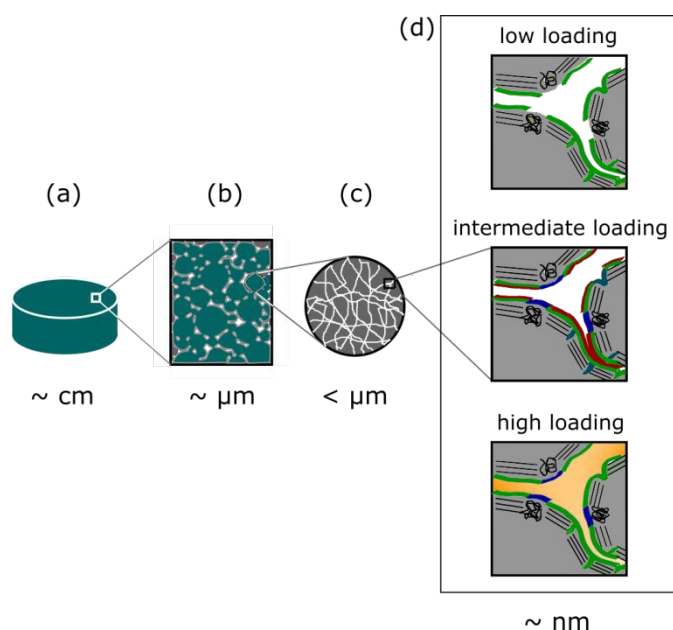


Figure 4: Schematic representation of different ionic liquid loading regimes in porous carbon black. (a) Pellet. (b) Excerpt of pellet, consisting of individual beads. (c) Individual highly porous bead. (d) Configuration of RTIL on pore level for different loading fractions. The stacks represent graphitic carbon and the clusters represent amorphous regions. RTIL with different mobility regimes are colour coded: (green) RTIL on graphitic planes; (blue) RTIL on amorphous carbon; (red) second RTIL layer on graphite; (teal) RTIL in cracks and cleavages between graphite crystallites; (yellow) bulk-like RTIL.